CURRENT AND FUTURE GENERATION

Thorium

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- Thorium is more abundant in nature than uranium.
- It is fertile rather than fissile, and can only be used as a fuel in conjunction with a fissile material such as recycled plutonium.
- Thorium fuels can breed fissile uranium-233 to be used in various kinds of nuclear reactors.
- Molten salt reactors are well suited to thorium fuel, as normal fuel fabrication is avoided.

The use of thorium as a new primary energy source has been a tantalizing prospect for many years. Extracting its latent energy value in a cost-effective manner remains a challenge, and will require considerable R&D investment. This is occurring preeminently in China, with modest US support.

Nature and sources of thorium

Thorium is a naturally-occurring, slightly radioactive metal discovered in 1828 by the Swedish chemist Jons Jakob Berzelius, who named it after Thor, the Norse god of thunder. It is found in small amounts in most rocks and soils, where it is about three times more abundant than uranium. Soil contains an average of around 6 parts per million (ppm) of thorium. Thorium is very insoluble, which is why it is plentiful in sands but not in seawater, in contrast to uranium.

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Thorium exists in nature in a single isotopic form – Th-232 – which decays very slowly (its half-life is about three times the age of the Earth). The decay chains of natural thorium and uranium give rise to minute traces of Th-228, Th-230 and Th-234, but the presence of these in mass terms is negligible. It decays eventually to lead-208.

When pure, thorium is a silvery white metal that retains its lustre for several months. However, when it is contaminated with the oxide, thorium slowly tarnishes in air, becoming grey and eventually black. When heated in air, thorium metal ignites and burns brilliantly with a white light. Thorium oxide (ThO_2), also called thoria, has one of the highest melting points of all oxides (3300°C) and so it has found applications in light bulb elements, lantern mantles, arc-light lamps, welding electrodes and heat-resistant ceramics. Glass containing thorium oxide has both a high refractive index and wavelength dispersion, and is used in high quality lenses for cameras and scientific instruments.

Thorium oxide (ThO_2) is relatively inert and does not oxidize further, unlike UO_2. It has higher thermal conductivity and lower thermal expansion than UO_2, as well as a much higher melting point. In nuclear fuel, fission gas release is much lower than in UO_2.

The most common source of thorium is the rare earth phosphate mineral, monazite, which contains up to about 12% thorium phosphate, but 6-7% on average. Monazite is found in igneous and other rocks but the richest concentrations are in placer deposits, concentrated by wave and current action with other heavy minerals. World monazite resources are estimated to be about 16 million tonnes, 12 Mt of which are in heavy mineral sands deposits on the south and east coasts of India. There are substantial deposits in several other countries (see Table below). Thorium recovery from monazite usually involves leaching with sodium hydroxide at 140°C followed by a complex process to precipitate pure ThO_2. Thorite ($ThSiO_4$) is another common thorium mineral. A large vein deposit of thorium and rare earth metals is in Idaho.

The International Atomic Energy Agency (IAEA) and the OECD Nuclear Energy Agency (NEA) joint publication *Uranium 2016: Resources, Production and Demand* (often referred to as the *Red Book*) gives a figure of 6.2 million tonnes of total known and estimated resources (the 2018 edition of the same publication did not provide estimates of thorium resources). Data for reasonably assured and inferred resources recoverable at a cost of $80/kg Th or less are given in the table below, excluding some less-certain Asian figures. Some of the figures are based on assumptions and surrogate data for mineral sands (monazite x assumed Th content),

not direct geological data in the same way as most mineral resources.

Estimated world thorium resources[1]

Country	Tonnes
India	846,000
Brazil	632,000
Australia	595,000
USA	595,000
Egypt	380,000
Turkey	374,000
Venezuela	300,000
Canada	172,000
Russia	155,000
South Africa	148,000
China	100,000
Norway	87,000
Greenland	86,000
Finland	60,000
Sweden	50,000
Kazakhstan	50,000
Other countries	1,725,000
World total	**6,355,000**

Source: OECD NEA & IAEA, *Uranium 2016: Resources, Production and Demand* ('Red Book')[1], using the lower figures of any range.

There is no international or standard classification for thorium resources and identified thorium resources do not have the same meaning in terms of classification as identified uranium resources. Thorium is not a primary exploration target and resources are estimated in relation to uranium and rare earths resources.

Monazite is extracted in India, Brazil, Vietnam and Malaysia, probably less than 10,000 t/yr, but without commercial rare earth recovery, thorium production is not economic at present. Chinese

production is unknown. The 2014 'Red Book' suggested that extraction of thorium as a by-product of rare earth elements (REE) recovery from monazite seems to be the most feasible source of thorium production at this time.

Thorium as a nuclear fuel

Thorium (Th-232) is not itself fissile and so is not directly usable in a thermal neutron reactor. However, it is 'fertile' and upon absorbing a neutron will transmute to uranium-233 (U-233)[a], which is an excellent fissile fuel material[b]. In this regard it is similar to uranium-238 (which transmutes to plutonium-239). All thorium fuel concepts therefore require that Th-232 is first irradiated in a reactor to provide the necessary neutron dosing to produce protactinium-233. The Pa-233 that is produced can either be chemically separated from the parent thorium fuel and the decay product U-233 then recycled into new fuel, or the U-233 may be usable 'in-situ' in the same fuel form, especially in molten salt reactors (MSRs).

Thorium fuels therefore need a fissile material as a 'driver' so that a chain reaction (and thus supply of surplus neutrons) can be maintained. The only fissile driver options are U-233, U-235 or Pu-239. (None of these is easy to supply)

It is possible – but quite difficult – to design thorium fuels that produce more U-233 in thermal reactors than the fissile material they consume (this is referred to as having a fissile conversion ratio of more than 1.0 and is also called breeding). Thermal breeding with thorium requires that the neutron economy in the reactor has to be very good (ie, there must be low neutron loss through escape or parasitic absorption). The possibility to breed fissile material in slow neutron systems is a unique feature for thorium-based fuels and is not possible with uranium fuels.

Another distinct option for using thorium is as a 'fertile matrix' for fuels containing plutonium that serves as the fissile driver while being consumed (and even other transuranic elements like americium). Mixed thorium-plutonium oxide (Th-Pu MOX) fuel is an analog of current uranium-MOX fuel, but no new plutonium is produced from the thorium component, unlike for uranium fuels in U-Pu MOX fuel, and so the level of net consumption of plutonium is high. Production of all actinides is lower than with conventional fuel, and negative reactivity coefficient is enhanced compared with U-Pu MOX fuel.

In fresh thorium fuel, all of the fissions (thus power and neutrons) derive from the driver component. As the fuel operates the U-233 content gradually increases and it contributes more and more to the power output of the fuel. The ultimate energy output from U-233 (and hence indirectly thorium) depends on numerous fuel design parameters, including: fuel burn-up attained, fuel arrangement, neutron energy spectrum and neutron flux (affecting the intermediate product protactinium-233, which is a neutron absorber). The fission of a U-233 nucleus releases about the same amount of energy (200 MeV) as that of U-235.

An important principle in the design of thorium fuel systems is that of heterogeneous fuel arrangement in which a high fissile (and therefore higher power) fuel zone called the seed region is physically separated from the fertile (low or zero power) thorium part of the fuel – often called the blanket. Such an arrangement is far better for supplying surplus neutrons to thorium nuclei so they can convert to fissile U-233, in fact all thermal breeding fuel designs are heterogeneous. This principle applies to all the thorium-capable reactor systems.

Th-232 is fissionable with fast neutrons of over 1 MeV energy. It could therefore be used in fast molten salt and other Gen IV reactors with uranium or plutonium fuel to initiate fission. However, Th-232 fast fissions only one tenth as well as U-238, so there is no particular reason for using thorium in fast reactors, given the huge amount of depleted uranium awaiting use.

In Norway, Thor Energy is developing and testing a thorium-bearing fuel for use in existing nuclear power plants. Fuel rods containing thorium additive (Th-Add) and also thorium MOX (with Pu) fuel rods were tested in a five-year irradiation trial that started in April 2013 at the Halden test reactor. The company is working towards obtaining regulatory approval for the commercial production and use of Th-Add fuel. In February 2018 a third batch of Th-MOX fuel pellets commenced testing. This fuel is promoted as a means to improve power profiles within commercial reactors.

Reactors able to use thorium

There are seven types of reactor into which thorium can be introduced as a nuclear fuel. The first five of these have all entered into operational service at some point. The last two are still conceptual:

Heavy Water Reactors (PHWRs): These are well suited for thorium fuels due to their combination of: (i) excellent neutron economy (their low parasitic neutron absorption means more neutrons can be absorbed by thorium to produce useful U-233), (ii) slightly faster average neutron energy which favours conversion to U-233, (iii) flexible on-line refueling capability. Furthermore, heavy water reactors (especially CANDU) are well established and widely-deployed commercial technology for which there is extensive licensing experience.

There is potential application to Enhanced Candu 6 (EC6) and ACR-1000 reactors fueled with 5% plutonium (reactor grade) plus thorium. In the closed fuel cycle, the driver fuel required for starting off is progressively replaced with recycled U-233, so that on reaching equilibrium 80% of the energy comes from thorium. Fissile drive fuel could be LEU, plutonium, or recycled uranium from LWR. Fleets of PHWRs with near-self-sufficient equilibrium thorium fuel cycles could be supported by a few fast breeder reactors to provide plutonium.

High-Temperature Gas-Cooled Reactors (HTRs): These are well suited for thorium-based fuels in the form of robust 'TRISO' coated particles of thorium mixed with plutonium or enriched uranium, coated with pyrolytic carbon and silicon carbide layers which retain fission gases. The fuel particles are embedded in a graphite matrix that is very stable at high temperatures. Such fuels can be irradiated for very long periods and thus deeply burn their original fissile charge. Thorium fuels can be designed for both 'pebble bed' and 'prismatic' types of HTR reactors.

Boiling (Light) Water Reactors (BWRs): BWR fuel assemblies can be flexibly designed in terms of rods with varying compositions (fissile content), and structural features enabling the fuel to experience more or less moderation (eg, half-length fuel rods). This design flexibility is very good for being able to come up with suitable heterogeneous arrangements and create well-optimised thorium fuels. So it is possible, for example, to design thorium-plutonium BWR fuels that are tailored for 'burning' surplus plutonium. And importantly, BWRs are a well-understood and licensed reactor type.

Pressurized (Light) Water Reactors (PWRs): Viable thorium fuels can be designed for a PWR, though with less flexibility than for BWRs. Fuel needs to be in heterogeneous arrangements in order to achieve satisfactory fuel burn-up. It is not possible to design viable thorium-based PWR fuels that convert significant amounts of U-233. Even though PWRs are not the perfect reactor in which to use

thorium, they are the industry workhorse and there is a lot of PWR licensing experience. They are a viable early-entry thorium platform.

Fast Neutron Reactors (FNRs): Thorium can serve as a fuel component for reactors operating with a fast neutron spectrum – in which a wider range of heavy nuclides are fissionable and may potentially drive a thorium fuel. There is, however, no relative advantage in using thorium instead of depleted uranium (DU) as a fertile fuel matrix in these reactor systems due to a higher fast-fission rate for U-238 and the fission contribution from residual U-235 in this material. Also, there is a huge amount of surplus DU available for use when more FNRs are commercially available, so thorium has little or no competitive edge in these systems.

Molten Salt Reactors (MSRs): These reactors are still at the design stage but are likely to be very well suited for using thorium as a fuel. The unique fluid fuel can incorporate thorium and uranium (U-233 and/or U-235) fluorides as part of a salt mixture that melts in the range 400-700°C, and this liquid serves as both heat transfer fluid and the matrix for the fissioning fuel. The fluid circulates through a core region and then through a chemical processing circuit that removes various fission products (poisons) and/or the valuable U-233. The level of moderation is given by the amount of graphite built into the core. Certain MSR designs[c] will be designed specifically for thorium fuels to produce useful amounts of U-233.

Accelerator Driven Reactors (ADS): The sub-critical ADS system is an unconventional nuclear fission energy concept that is potentially 'thorium capable'. Spallation neutrons are produced[d] when high-energy protons from an accelerator strike a heavy target like lead. These neutrons are directed at a region containing a thorium fuel, eg, Th-plutonium which reacts to produce heat as in a conventional reactor. The system remains subcritical ie, unable to sustain a chain reaction without the proton beam. Difficulties lie with the reliability of high-energy accelerators and also with economics due to their high power consumption. (See also information page on Accelerator-Driven Nuclear Energy.)

A key finding from thorium fuel studies to date is that it is not economically viable to use low-enriched uranium (LEU – with a U-235 content of up to 20%) as a fissile driver with thorium fuels, unless the fuel burn-up can be taken to very high levels – well beyond those currently attainable in LWRs with zirconium cladding.

With regard to proliferation significance, thorium-based power reactor fuels would be a poor source for fissile material usable in

the illicit manufacture of an explosive device. U-233 contained in spent thorium fuel contains U-232 which decays to produce very radioactive daughter nuclides and these create a strong gamma radiation field. This confers proliferation resistance by creating significant handling problems and by greatly boosting the detectability (traceability) and ability to safeguard this material.

Prior thorium-fuelled electricity generation

There have been several significant demonstrations of the use of thorium-based fuels to generate electricity in several reactor types. Many of these early trials were able to use high-enriched uranium (HEU) as the fissile 'driver' component, and this would not be considered today.

The 300 MWe Thorium High Temperature Reactor (**THTR)** at Hamm-Uentrop in Germany operated with thorium-HEU fuel between 1983 and 1989, when it was shut down due to technical problems. Over half of its 674,000 pebbles contained Th-HEU fuel particles (the rest comprised graphite moderator and some neutron absorbers). These were continuously moved through the reactor as it operated, and on average each fuel pebble passed six times through the core.

The 40 MWe **Peach Bottom HTR** in the USA was a demonstration thorium-fuelled reactor that ran from 1967-74.[2] It used a thorium-HEU fuel in the form of microspheres of mixed thorium-uranium carbide coated with pyrolytic carbon. These were embedded in annular graphite segments (not pebbles). This reactor produced 33 billion kWh over 1349 equivalent full-power days with a capacity factor of 74%.

The 330 MWe **Fort St Vrain HTR** in Colorado, USA, was a larger-scale commercial successor to the Peach Bottom reactor and ran from 1976-89. It also used thorium-HEU fuel in the form of microspheres of mixed thorium-uranium carbide coated with silicon oxide and pyrolytic carbon to retain fission products. These were embedded in graphite 'compacts' that were arranged in hexagonal columns ('prisms'). Almost 25 tonnes of thorium was used in fuel for the reactor, much of which attained a burn-up of about 170 GWd/t.

A unique thorium-fuelled light water breeder reactor operated from 1977 to 1982 at **Shippingport** in the USA[3] – it used uranium-233 as the fissile driver in special fuel assemblies that had movable 'seed' regions which allowed the level of neutron moderation to be gradually increased as the fuel aged[e]. The reactor core was housed

in a reconfigured early PWR. It operated with a power output of 60 MWe (236 MWt) and an availability factor of 86% producing over 2.1 billion kWh. Post-operation inspections revealed that 1.39% more fissile fuel was present at the end of core life, proving that breeding had occurred. A 2007 NRC report quotes a breeding ratio of 1.01. Chemically reprocessing the fuel was not attempted.

Indian heavy water reactors (**PHWRs**) have for a long time used thorium-bearing fuel bundles for power flattening in some fuel channels – especially in initial cores when special reactivity control measures are needed.

Thorium energy R&D – past & present

Research into the use of thorium as a nuclear fuel has been taking place for over 50 years, though with much less intensity than that for uranium or uranium-plutonium fuels. Basic development work has been conducted in Germany, India, Canada, Japan, China, Netherlands, Belgium, Norway, Russia, Brazil, the UK & the USA. Test irradiations have been conducted on a number of different thorium-based fuel forms.

Noteworthy studies and experiments involving thorium fuel include:

Heavy water reactors: Thorium-based fuels for the 'Candu' PHWR system have been designed and tested in Canada at AECL's Chalk River Laboratories for more than 50 years, including the irradiation of ThO_2-based fuels to burn-ups to 47 GWd/t. Dozens of test irradiations have been performed on fuels including: mixed ThO_2-UO_2, (both LEU and HEU), and mixed ThO_2-PuO_2, (both reactor- and weapons-grade). The NRX, NRU and WR-1 reactors were used, NRU most recently. R&D into thorium fuel use in CANDU reactors continues to be pursued by Canadian and Chinese groups as part of joint studies looking at a wide range of fuel cycle options involving China's Qinshan Phase III PHWR units. Eight ThO_2-based fuel pins have been successfully irradiated in the middle of a LEU Candu fuel bundle with low-enriched uranium. The fuels have performed well in terms of their material properties.

Closed thorium fuel cycles have been designed[4] in which PHWRs play a key role due to their fuelling flexibility: thoria-based HWR fuels can incorporate recycled U-233, residual plutonium and uranium from used LWR fuel, and also minor actinide components in waste-reduction strategies. In the closed cycle, the driver fuel required for starting off is progressively replaced with recycled U-

233, so that an ever-increasing energy share in the fuel comes from the thorium component. AECL had a Thoria Roadmap R&D project.

In July 2009 a second phase agreement was signed among AECL, the Third Qinshan Nuclear Power Company (TQNPC), China North Nuclear Fuel Corporation and the Nuclear Power Institute of China to jointly develop and demonstrate the use of thorium fuel and to study the commercial and technical feasibility of its full-scale use in Candu units such as at Qinshan. An expert panel appointed by CNNC unanimously recommended that China consider building two new Candu units to take advantage of the design's unique capabilities in utilizing alternative fuels. It confirmed that thorium use in the Enhanced Candu 6 reactor design is “technically practical and feasible”, and cited the design’s “enhanced safety and good economics” as reasons it could be deployed in China in the near term.

India’s nuclear developers have designed an Advanced Heavy Water Reactor (AHWR) specifically as a means for ‘burning’ thorium – this will be the final phase of their three-phase nuclear energy infrastructure plan (see below). The reactor will operate with a power of 300 MWe using thorium-plutonium or thorium-U-233 seed fuel in mixed oxide form. It is heavy water moderated (& light water cooled) and will eventually be capable of self-sustaining U-233 production. In each assembly 30 of the fuel pins will be Th-U-233 oxide, arranged in concentric rings. About 75% of the power will come from the thorium. Construction of the pilot AHWR was envisaged in the 12th plan period to 2017, for operation about 2022. As of 2020, however, no site or construction schedule for the demonstration unit has been announced.

For export, India has also designed an AHWR300-LEU which uses low-enriched uranium as well thorium in fuel, dispensing with plutonium input. About 39% of the power will come from thorium (via in situ conversion to U-233, c. two-thirds in AHWR), and burn-up will be 64 GWd/t. While closed fuel cycle is possible, this is not required or envisaged, and the used fuel, with about 8% fissile isotopes can be used in light water reactors. (See also information page on India).

High-temperature gas-cooled reactors: Thorium fuel was used in HTRs prior to the successful demonstration reactors described above. The UK operated the 20 MWth Dragon HTR from 1964 to 1973 for 741 full power days. Dragon was run as an OECD/Euratom cooperation project, involving Austria, Denmark, Sweden, Norway and Switzerland in addition to the UK. This reactor used thorium-HEU fuel elements in a 'breed and feed' mode

in which the U-233 formed during operation replaced the consumption of U-235 at about the same rate. The fuel comprised small particles of uranium oxide (1 mm diameter) coated with silicon carbide and pyrolytic carbon which proved capable of maintaining a high degree of fission product containment at high temperatures and for high burn-ups. The particles were consolidated into 45mm long elements, which could be left in the reactor for about six years.

Germany operated the Atom Versuchs Reaktor (AVR) at Jülich for over 750 weeks between 1967 and 1988. This was a small pebble bed reactor that operated at 15 MWe, mainly with thorium-HEU fuel. About 1360 kg of thorium was used in some 100,000 pebbles. Burn-ups of 150 GWd/t were achieved.

Pebble bed reactor development builds on German work with the AVR and THTR and is under development in China (HTR-10, and HTR-PM).

Light water reactors: The feasibility of using thorium fuels in a PWR was studied in considerable detail during a collaborative project between Germany and Brazil in the 1980s[5]. The vision was to design fuel strategies that used materials effectively – recycling of plutonium and U-233 was seen to be logical. The study showed that appreciable conversion to U-233 could be obtained with various thorium fuels, and that useful uranium savings could be achieved. The program terminated in 1988 for non-technical reasons. It did not reach its later stages which would have involved trial irradiations of thorium-plutonium fuels in the Angra-1 PWR in Brazil, although preliminary Th-fuel irradiation experiments were performed in Germany. Most findings from this study remain relevant today.

Thorium-plutonium oxide (Th-MOX) fuels for LWRs are being developed by Norwegian proponents (see above) with a view that these are the most readily achievable option for tapping energy from thorium. This is because such fuel is usable in existing reactors (with minimal modification) using existing uranium-MOX technology and licensing experience.

Various groups are evaluating the option of using thorium fuels in an advanced reduced-moderation BWR (RBWR). This reactor platform, designed by Hitachi Ltd and JAEA, should be well suited for achieving high U-233 conversion factors from thorium due to its epithermal neutron spectrum. High levels of actinide destruction may also be achieved in carefully designed thorium fuels in these conditions. The RBWR is based on the ABWR architecture but has

a shorter, flatter pancake-shaped core and a tight hexagonal fuel lattice to ensure sufficient fast neutron leakage and a negative void reactivity coefficient.

The so-called Radkowsky Thorium Reactor design is based on a heterogeneous 'seed & blanket' thorium fuel concept, tailored for Russian-type LWRs (VVERs)[6]. Enriched uranium (20% U-235) or plutonium is used in a seed region at the centre of a fuel assembly, with this fuel being in a unique metallic form. The central seed portion is demountable from the blanket material which remains in the reactor for nine years[f], but the centre seed portion is burned for only three years (as in a normal VVER). Design of the seed fuel rods in the centre portion draws on experience of Russian naval reactors.

The European Framework Program has supported a number of relevant research activities into thorium fuel use in LWRs. Three distinct trial irradiations have been performed on thorium-plutonium fuels, including a test pin loaded in the Obrigheim PWR over 2002-06 during which it achieved about 38 GWd/t burnup.

A small amount of thorium-plutonium fuel was irradiated in the 60 MWe Lingen BWR in Germany in the early 1970s. The fuel contained 2.6 % of high fissile-grade plutonium (86% Pu-239) and the fuel achieved about 20 GWd/t burnup. The experiment was not representative of commercial fuel, however the experiment allowed for fundamental data collection and benchmarking of codes for this fuel material.

In 2020 a consortium was formed to develop Advanced Nuclear Energy for Enriched Life (ANEEL) fuel, a mixture of high-assay low-enriched uranium (HALEU) and thorium. This can be used in BWR and PWRs but seems best suited for PHWR reactors such as CANDU where fuel would remain in the core for about eight times longer than usual. It has a high burn-up to 55 GWd/t. The consortium comprises the DOE's Idaho National Laboratory and the Nuclear Engineering & Science Center at Texas A&M University with Clean Core Thorium Energy (CCTE).

Molten salt reactors: In the 1960s the Oak Ridge National Laboratory (USA) designed and built a demonstration MSR using U-233 as the main fissile driver in its second campaign. The reactor ran over 1965-69 at powers up to 7.4 MWt. The lithium-beryllium salt worked at 600-700°C and ambient pressure. The R&D program demonstrated the feasibility of this system and highlighted some unique corrosion and safety issues that would need to be addressed if constructing a larger pilot MSR.

There is significant renewed interest in developing thorium-fuelled MSRs. Projects are (or have recently been) underway in China, Japan, Russia, France and the USA. It is notable that the MSR is one of the six 'Generation IV' reactor designs selected as worthy of further development (see information page on Generation IV Nuclear Reactors).

The thorium-fuelled MSR variant is sometimes referred to as the **Liquid Fluoride Thorium Reactor** (LFTR), utilizing U-233 which has been bred in a liquid thorium salt blanket.[g]

Safety is achieved with a freeze plug which if power is cut allows the fuel to drain into subcritical geometry in a catch basin. There is also a negative temperature coefficient of reactivity due to expansion of the fuel.

The China Academy of Sciences in January 2011 launched an R&D program on LFTR, known there as the thorium-breeding molten salt reactor (Th-MSR or TMSR), and claimed to have the world's largest national effort on it, hoping to obtain full intellectual property rights on the technology. The TMSR Research Centre has a 5 MWe MSR prototype under construction at Shanghai Institute of Applied Physics (SINAP, under the Academy).

SINAP has two streams of MSR development – solid fuel (TRISO in pebbles or prisms/blocks) with once-through fuel cycle, and liquid fuel (dissolved in FLiBe coolant) with reprocessing and recycle. A third stream of fast reactors to consume actinides from LWRs is planned.

- The TMSR-SF (solid fuel) stream has only partial utilization of thorium, relying on some breeding as with U-238, and needing fissile uranium input as well. SINAP aims at a 2 MW pilot plant initially, and a 100 MWt demonstration pebble bed plant with open fuel cycle by about 2030. TRISO particles will be with both low-enriched uranium and thorium, separately.

- The TMSR-LF (liquid fuel) stream claims full closed Th-U fuel cycle with breeding of U-233 and much better sustainability but greater technical difficulty. SINAP aims for a 10 MWt pilot plant by 2030 and a 100 MWt demonstration plant by 2040.

- A TMSR-LF fast reactor optimized for burning minor actinides is to follow.

SINAP sees molten salt fuel being superior to the TRISO fuel in effectively unlimited burn-up, less waste, and lower fabricating

cost, but achieving lower temperatures (600°C+) than the TRISO fuel reactors (1200°C+). Near-term goals include preparing nuclear-grade ThF_4 and ThO_2 and testing them in a MSR. The US Department of Energy (especially Oak Ridge NL) is collaborating with the Academy on the program, which had a start-up budget of $350 million.

However, the primary reason that American researchers and the China Academy of Sciences/ SINAP are working on solid fuel, salt-cooled reactor technology is that it is a realistic first step. The technical difficulty of using molten salts is significantly lower when they do not have the very high activity levels associated with them bearing the dissolved fuels and wastes. The experience gained with component design, operation, and maintenance with clean salts makes it much easier then to move on and consider the use of liquid fuels, while gaining several key advantages from the ability to operate reactors at low pressure and deliver higher temperatures.

Accelerator-driven reactors: A number of groups have investigated how a thorium-fuelled accelerator-driven reactor (ADS) may work and appear. Perhaps most notable is the 'ADTR' design patented by a UK group. This reactor operates very close to criticality and therefore requires a relatively small proton beam to drive the spallation neutron source. Earlier proposals for ADS reactors required high-energy and high-current proton beams which are energy-intensive to produce, and for which operational reliability is a problem.

Research reactor 'Kamini': India has been operating a low-power U-233 fuelled reactor at Kalpakkam since 1996 – this is a 30 kWth experimental facility using U-233 in aluminium plates (a typical fuel-form for research reactors). Kamini is water cooled with a beryllia neutron reflector. The total mass of U-233 in the core is around 600 grams. It is noteworthy for being the only U-233 fuelled reactor in the world, though it does not in itself directly support thorium fuel R&D. The reactor is adjacent to the 40 MWt Fast Breeder Test Reactor in which ThO_2 is irradiated, producing the U-233 for Kamini.

Aqueous homogeneous reactor: An aqueous homogenous suspension reactor operated over 1974-77 in the Netherlands at 1 MWth using thorium plus HEU oxide pellets. The thorium-HEU fuel was circulated in solution with continuous reprocessing outside the core to remove fission products, resulting in a high conversion rate to U-233.

Developing a thorium-based fuel cycle

Thorium fuel cycles offer attractive features, including lower levels of waste generation, less transuranic elements in that waste, and providing a diversification option for nuclear fuel supply. Also, the use of thorium in most reactor types leads to extra safety margins. Despite these merits, the commercialization of thorium fuels faces some significant hurdles in terms of building an economic case to undertake the necessary development work.

A great deal of testing, analysis and licensing and qualification work is required before any thorium fuel can enter into service. This is expensive and will not eventuate without a clear business case and government support. Also, uranium is abundant and cheap and forms only a small part of the cost of nuclear electricity generation, so there are no real incentives for investment in a new fuel type that may save uranium resources.

Other impediments to the development of thorium fuel cycle are the higher cost of fuel fabrication and the cost of reprocessing to provide the fissile plutonium driver material. The high cost of fuel fabrication (for solid fuel) is due partly to the high level of radioactivity that builds up in U-233 chemically separated from the irradiated thorium fuel. Separated U-233 is always contaminated with traces of U-232 which decays (with a 69-year half-life) to daughter nuclides such as thallium-208 that are high-energy gamma emitters. Although this confers proliferation resistance to the fuel cycle by making U-233 hard to handle and easy to detect, it results in increased costs. There are similar problems in recycling thorium itself due to highly radioactive Th-228 (an alpha emitter with two-year half life) present. Some of these problems are overcome in the LFTR or other molten salt reactor and fuel cycle designs, rather than solid fuel.

Particularly in a molten salt reactor, the equilibrium fuel cycle is expected to have relatively low radiotoxicity, being fission products only plus short-lived Pa-233, without transuranics. These are continually removed in on-line reprocessing, though this is more complex than for the uranium-plutonium fuel cycle.

Nevertheless, the thorium fuel cycle offers energy security benefits in the long-term – due to its potential for being a self-sustaining fuel without the need for fast neutron reactors. It is therefore an important and potentially viable technology that seems able to contribute to building credible, long-term nuclear energy scenarios.

India's plans for thorium cycle

With huge resources of easily-accessible thorium and relatively little uranium, India has made utilization of thorium for large-scale energy production a major goal in its nuclear power programme, utilising a three-stage concept first proposed at the University of Chicago in 1944:

1. Pressurized heavy water reactors (PHWRs) and light water reactors fuelled by natural uranium producing plutonium that is separated for use in fuels in its fast reactors and indigenous advanced heavy water reactors.
2. Fast breeder reactors (FBRs) will use plutonium-based fuel to extend their plutonium inventory. The blanket around the core will have uranium as well as thorium, so that further plutonium (particularly Pu-239) is produced as well as U-233.
3. Advanced heavy water reactors (AHWRs) will burn thorium-plutonium fuels in such a manner that breeds U-233 which can eventually be used as a self-sustaining fissile driver for a fleet of breeding AHWRs. The final core of the Shippingport reactor in the USA demonstrated this.

In all of these stages, used fuel needs to be reprocessed to recover fissile materials for recycling.

India is focusing and prioritizing the construction and commissioning of its fleet of 500 MWe sodium-cooled fast reactors in which it will breed the required plutonium which is the key to unlocking the energy potential of thorium in its advanced heavy water reactors. This will take another 15-20 years, and so it will still be some time before India is using thorium energy to any extent. The 500 MWe prototype FBR under construction in Kalpakkam was expected to start up in 2014, but 2021 is now the target date.

In 2009, despite the relaxation of trade restrictions on uranium, India reaffirmed its intention to proceed with developing the thorium cycle.

Weapons and non-proliferation

The thorium fuel cycle is sometimes promoted as having excellent non-proliferation credentials. This is true, but some history and physics bears noting.

The USA produced about 2 tonnes of U-233 from thorium during the 'Cold War', at various levels of chemical and isotopic purity, in plutonium production reactors. It is possible to use U-233 in a nuclear weapon, and in 1955 the USA detonated a device with a plutonium-U-233 composite pit, in Operation Teapot. The explosive yield was less than anticipated, at 22 kilotons. In 1998 India detonated a very small device based on U-233 called Shakti V. However, the production of U-233 inevitably also yields U-232 which is a strong gamma-emitter, as are some decay products such as thallium-208 ('thorium C'), making the material extremely difficult to handle and also easy to detect.

U-233 is classified by the IAEA in the same category as high enriched uranium (HEU), with a significant quantity in terms of safeguards defined as 8 kg, compared with 32 kg for HEU.

Notes & references

Notes

a. Neutron absorption by Th-232 produces Th-233 which beta-decays (with a half-life of about 22 minutes) to protactinium-233 (Pa-233) – and this decays to U-233 by further beta decay (with a half-life of 27 days). Some of the bred-in U-233 is converted to U-234 by further neutron absorption. U-234 is an unwanted parasitic neutron absorber. It converts to fissile U-235 (the naturally occuring fissile isotope of uranium) and this somewhat compensates for this neutronic penalty. In fuel cycles involving the multi-recycle of thorium-U-233 fuels, the build up of U-234 can be appreciable. [Back]

b. A U-233 nucleus yields more neutrons, on average, when it fissions (splits) than either a uranium-235 or plutonium-239 nucleus. In other words, for every thermal neutron absorbed in a U-233 fuel there are a greater number of neutrons produced and released into the surrounding fuel. This gives better neutron economy in the reactor system.. [Back]

c. MSRs using thorium will likely have a distinct 'blanket' circuit which is optimized for producing U-233 from dissolved thorium.

Neutron moderation is tailored by the amount of graphite in the core (aiming for an epithermal spectrum). This uranium can be selectively removed as uranium hexafluoride (UF6) by bubbling fluorine gas through the salt. After conversion it can be directed to the core as fissile fuel. [Back]

d. Spallation is the process where nucleons are ejected from a heavy nucleus being hit by a high energy particle. In this case, a high-enery proton beam directed at a heavy target expels a number of spallation particles, including neutrons. [Back]

e. The core of the Shippingport demonstration LWBR consisted of an array of seed and blanket modules surrounded by an outer reflector region. In the seed and blanket regions, the fuel pellets contained a mixture of thorium-232 oxide (ThO_2) and uranium oxide (UO_2) that was over 98% enriched in U-233. The proportion of UO_2 was around 5-6% in the seed region, and about 1.5-3% in the blanket region. The reflector region contained only thorium oxide at the beginning of the core life. [Back]

f. Blanket fuel is designed to reach 100 GWd/t burn-up. Together, the seed and blanket have the same geometry as a normal VVER-100 fuel assembly (331 rods in a hexagonal array 235 mm wide). [Back]

g. The molten salt in the core circuit consists of lithium, beryllium and fissile U-233 fluorides (FLiBe with uranium). It operates at some 700°C and circulates at low pressure within a graphite structure that serves as a moderator and neutron reflector. Most fission products dissolve or suspend in the salt and some of these are removed progressively in an adjacent on-line radiochemical processing unit. Actinides are less-readily formed than in fuel with atomic mass greater than 235. The blanket circuit contains a significant amount of thorium tetrafluoride in the molten Li-Be fluoride salt. Newly-formed U-233 forms soluble uranium tetrafluoride (UF_4), which is converted to gaseous uranium hexafluoride (UF_6) by bubbling fluorine gas through the salt (which does not chemically affect the less-reactive thorium tetrafluoride). The volatile uranium hexafluoride is captured, reduced back to soluble UF_4 by hydrogen gas, and finally is directed to the core to serve as fissile fuel. Protactinium – a neutron absorber – is not a major problem in the blanket salt. [Back]

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